AI in Buying, Merchandising and E-commerce

WHY THIS MODULE MATTERS

Buying, merchandising and e-commerce decisions increasingly rely on algorithms. This module ensures human judgement stays in the loop where it matters most.

7.1 Algorithms Now Shape What Customers Can Buy

Buying and merchandising decisions increasingly rest on algorithmic forecasts: what to order, in what quantities, at what price, and what to show first on a category page. These systems rarely make headlines, because their failures show up as margin erosion, stock imbalance or quietly narrowed customer choice rather than as public scandals. That makes them easier to under-govern — and, when they do surface publicly, they surface as a regulatory or consumer-protection problem rather than a technical one.

7.2 Recommendation Systems and Narrowed Choice

Recommendation engines optimise for what customers are likely to click or buy, based on what similar customers did before. Over time this creates a feedback loop: products the engine surfaces generate sales data, which reinforces their prominence, while products it does not surface generate none, which the model reads as low demand rather than low exposure. New, niche, adaptive and extended-size ranges are the most common casualties — not through any deliberate decision, but because the system never gave them enough exposure to prove themselves.

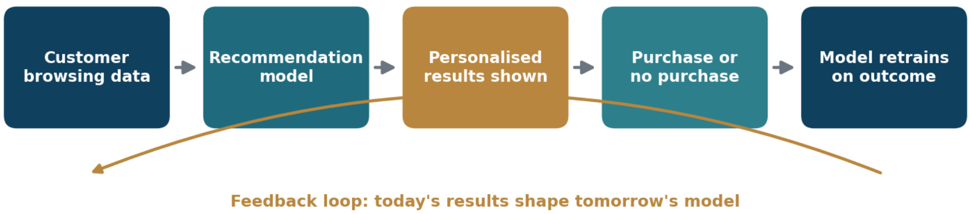

Figure 7.1 — How a recommendation feedback loop can suppress a product range without anyone deciding to.

- **Reserve exploration capacity** — deliberately surface a proportion of products the model is uncertain about, so new lines can generate real signal.
- **Monitor coverage, not just conversion** — track what share of the catalogue is ever surfaced, and to whom.
- **Check subgroup outcomes** — confirm that extended sizes, adaptive ranges and modest-wear lines are being surfaced to the customers actually looking for them.

- **Separate 'no demand' from 'no exposure'** — before cutting a range on the basis of weak sales, establish whether it was ever given a fair chance to sell.

7.3 Ranking, Search and Transparency Obligations

How products are ranked is not purely a commercial choice. Under UK consumer protection law, online retailers and marketplaces must disclose the main parameters determining how products are ranked in search results, and must make clear where placement has been paid for. An AI ranking model does not remove this obligation — if a retailer cannot explain in general terms why one product outranks another, it cannot meet the disclosure requirement.

This creates a practical governance requirement that teams often discover late: a ranking model needs to be explainable at least at the level of "these are the factors it weighs", even where the precise weighting is commercially sensitive. Buying a third-party ranking engine that cannot answer that question transfers a compliance problem into the business along with the software.

CASE STUDY — THE EU-WIDE ONLINE RETAIL SWEEP

In January 2023, the European Commission and national consumer protection authorities in the Consumer Protection Cooperation network published the results of a coordinated sweep of 399 online retail websites. Nearly 40% — 148 of the 399 sites — were found to be using at least one manipulative design practice: fake countdown timers pressuring customers toward a bogus deadline, interfaces designed to steer customers toward particular choices, or hidden information about the true nature of an offer.

The finding that matters most for this module is how ordinary these practices had become. These were not fringe operators; nearly four in ten mainstream retail sites carried at least one such pattern, in many cases implemented as a default feature of an off-the-shelf e-commerce or personalisation platform rather than as a deliberate decision by the retailer. The regulatory direction of travel since has been consistently toward tighter enforcement, and UK retailers should expect scrutiny of automated urgency and ranking practices to increase rather than ease.

7.4 Forecasting, Pricing and Human Judgement

Demand forecasting and dynamic pricing models are among the most commercially valuable AI systems in retail, and among the most prone to quiet automation bias. Buyers who would once have challenged an implausible number now frequently accept it, because it arrived from a system rather than a colleague. The failure mode is not that the model is wrong — models are often wrong — but that no one is positioned to notice.

- Require a stated reason when a buyer overrides a forecast, and equally when they accept one that contradicts their own experience.
- Track forecast accuracy by category, supplier and size curve, not only in aggregate.
- Set thresholds where an unusual recommendation triggers mandatory human review before action — for example, any proposed buy change beyond a set percentage.

- Keep dynamic pricing tied to legitimate factors — stock levels, season, competitor pricing — rather than inferred individual willingness to pay (see Module 6).
- Record who approved a significant automated recommendation, so accountability does not dissolve into ‘the system said so’.

7.5 Markdown, Returns and Fraud Models

Three further model types deserve specific governance attention because each can affect individual customers directly rather than only commercial outcomes.

- **Markdown optimisation** — generally low-risk, since it operates at product rather than person level, but should be checked to ensure discounting does not systematically concentrate in ranges serving particular customer groups.
- **Returns-abuse models** — higher-risk, because they profile individuals and can result in account restrictions. These edge toward Article 22 territory (Module 5) and need a human review step and an appeal route before any restriction is applied.
- **Payment fraud models** — highest-risk of the three, since a false positive blocks a legitimate customer from transacting. Monitor false-positive rates by customer segment, not just overall fraud caught.

IN PRACTICE

A retailer reviews its returns-abuse model and finds it flags customers who order multiple sizes and return the surplus — behaviour the brand actively encourages elsewhere in its own fit guidance, and which is most common among customers who find sizing least consistent. The rule is rewritten to exclude same-style multi-size orders before any accounts are restricted.

TRY THIS — YOU ARE THE MERCHANDISING LEAD

Scenario: your demand model recommends cutting the buy on your extended-size range by 60% for next season, citing weak sales history. A buyer notes the range was only ever shown on a filtered sub-page and rarely appeared in recommendations. Do you accept the recommendation? (Model answer: no — not as it stands. The weak sales history is at least partly an artefact of low exposure, not proven low demand: a textbook feedback loop from section 7.2. The right response is to challenge the input, not just the output — run a period of deliberate exposure to generate genuine demand signal before committing to a cut of that scale. Accepting the number uncritically is exactly the automation bias described in section 7.4, and cutting an extended-size range on flawed evidence also carries a reputational and potentially an equality dimension.)

KEY TAKEAWAYS

• Buying and merchandising AI is easy to under-govern because its failures show up as margin and choice erosion rather than public incidents.
• Recommendation feedback loops systematically disadvantage new, niche, adaptive and extended-size ranges by never giving them enough exposure to prove demand.
• Ranking transparency is a legal obligation, not just good practice — a model you cannot explain in general terms is a model you cannot disclose.
• The EU-wide sweep found manipulative patterns on nearly 40% of retail sites, often as platform defaults rather than deliberate choices — audit what your tools do by default.
• Automation bias is the core forecasting risk: the danger is not that the model is wrong, but that nobody is positioned to notice.
• Returns-abuse and fraud models profile individuals and need human review and an appeal route before any restriction is applied.

FURTHER READING

— Competition and Markets Authority — guidance on online choice architecture and ranking transparency
— European Commission — 2023 CPC coordinated sweep of online retail sites
— ICO — guidance on fairness and automated decision-making

MODULE 7 CHECK-IN QUIZ

Five questions to confirm your understanding before moving on. Answers are provided in the Answer Key at the end of the course.

1. Why is buying and merchandising AI easy to under-govern?

A. It is not used in most retailers
B. It never processes personal data
C. Regulators have exempted it
D. Its failures show up as margin and choice erosion rather than public incidents

2. What is the feedback loop problem in recommendation engines?

A. Products the engine surfaces generate sales data reinforcing their prominence, while unsurfaced products generate none, which reads as low demand
B. Models improve automatically over time with no intervention
C. Customers give too much explicit feedback
D. Engines recommend only the newest products

3. Which product ranges are most commonly disadvantaged by recommendation feedback loops?

A. Best-selling core lines
B. New, niche, adaptive and extended-size ranges
C. Discounted clearance stock
D. Own-brand basics

4. What is the core risk of automation bias in demand forecasting?

A. That models are frequently wrong

B. That forecasts take too long to produce

C. That nobody is positioned to notice when a model is wrong, because the number came from a system rather than a colleague

D. That buyers override forecasts too often

5. In the extended-size range exercise, why should the recommendation be challenged?

A. The model is always unreliable

B. Extended sizes are legally protected from range cuts

C. The buyer outranks the model

D. The weak sales history is at least partly an artefact of low exposure, not proven low demand